



CIVIL MAINT. Department
SIMHADRI

Note No :1

Ref:SMTTP/Civil Maint/2023-24/NTPCSIMH/691893

Date:30/11/2023(09:27:20)

Subject:LMI FOR INSPECTION OF CONCRETE CHIMNEY-REG

The LMI No. LMI/OGN/OPS/CIV/005, rev 3 dated 25.07.2012 for inspection of concrete chimney is reviewed and no changes are required to be made with respect to previous version.

Put up for approval of competent authority.

Submitted by:

प्रकाश चंद्र गुप्ता - वरिष्ठ प्रबंधक

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On:30/11/2023(09:27:20) **Ref:**SMTTP/Civil Maint/2023-24/NTPCSIMH/691893

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Note No:6

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Note No:7

Reviewed w.r.t OGN/OPS/CIV/005. Forwarded for approval.

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Note No:8

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Note No:9

Review / Comment from Lighting/Earthing/Lift groups of EMD may be done

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Note No:10

Reviewed w.r.t OGN/OPS/CIV/005. forwarded further for review

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Note No:11

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Note No:12

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Note No:13

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May please be approved

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CIVIL MAINT. Department
SIMHADRI

Ref:SMTPP/Civil Maint/2021-22/NTPCSIMH/610210

Date:06/12/2021(18:34:32)

Subject:LMI FOR INSPECTION OF CONCRETE CHIMNEY-REG

The LMI No. LMI/OGN/OPS/CIV/005, rev 3 dated 25.07.2012 for inspection of concrete chimney is reviewed and no changes are required to be made with respect to previous version.

Put up for approval of competent authority.

Submitted by:

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There are no changes envisaged in CC-OS document. Hence the proposal may be approved

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On:13/12/2021(11:54:01)

Reference files attached with this file :

The following LMI's has been reviewed and changes as mentioned below are required in respective LMI's with respect to their earlier revisions.

S.No.	LMI No.	LMI description	Clause no./Text	Clause no./changed text
1	LMI/OGN/OPS/CIV/00 5 Rev.03	Inspection of Chimney	-	-
2	LMI/OGN/OPS/CIV/00 6 Rev.01	Inspection of storm water drains in plant area	10 This document shall be reviewed by <u>GM(I/C)</u> on two yearly basis	10 This document shall be reviewed by <u>HOP Simhadri</u> on two yearly basis
3	LMI/OGN/OPS/CIV/00 7 Rev.04	Inspection and Maintenance of steel and concrete structures	11 This document shall be reviewed by <u>GM(I/C)</u> on two yearly basis	11 This document shall be reviewed by <u>HOP Simhadri</u> on two yearly basis
4	LMI/OGN/OPS/CIV/01 1 Rev.03	Inspection of Machine Foundations	11 This document shall be reviewed by <u>GM(I/C)</u> on three yearly basis	11 This document shall be reviewed by <u>HOP Simhadri</u> on two yearly basis
5	LMI/OGN/OPS/CIV/01 2 Rev.01	Inspection and Maintenance of Structure/Buildings in the vicinity of corrosive chemicals, Effluents and Fumes	11 This document shall be reviewed by <u>GM(I/C)</u> Simhadri on two years basis	11 This document shall be reviewed by <u>HOP Simhadri</u> on two years basis

The above changes have been incorporated in the respective LMI's.

Submitted for approval please.

J. Gopi Kiran
(J.Gopi Kiran)
DGM (O&M-Civil)

AGM (Civil, OFS)

GM (M)

GM (O&M)

CGM

21/01/2020

03.02.2020

09/02/2020

11/02/2020



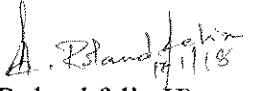
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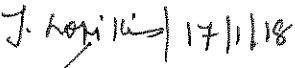
Date:17.01.2018

Sub: LMI for inspection of concrete chimney

The LMI No. LMI/OGN/OPS/CIV/005, rev-03 dtd. 25.07.2012 for inspection of concrete chimney is reviewed and no changes are required to be made with respect to previous version.

Put up for approval of competent authority.

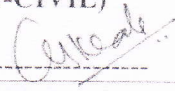
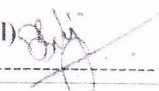

(Roland felix H)
Dy.Mgr (Plant Civil Maint.)


Sr.Manager (Plant Civil Maint.)


AGM (Civil) 17.1.18



LMI APPROVAL DOCUMENTS

<u>1.0 TITLE OF LMI</u>	INSPECTION OF CONCRETE CHIMNEY
<u>2.a) DOCUMENT CODE</u>	<u>LMI/OGN/OPS/CIV/005/</u>
<u>b) REVISION/ISSUE NO.&DATE</u>	<u>ISSUE-1; REV.-3.00; JULY'2012</u>
<u>3.ORIGINAL/REF.DOCUMENT</u>	<u>OGN/OPS/CIV/005</u>
<u>4.PRERPREPARED BY</u>	NAME: Shri.A.V.Satyanarayana. Sr.Mgr (O&M-CIVIL) SIGNATURE----- 
<u>5. CHECKED BY</u>	NAME: Shri.DSGSS.Babji AGM (RM & Civil) SIGNATURE----- 
<u>6.)REVIEWED BY</u>	NAME: Shri.A.V.S. Pandarinath AGM (MM.) SIGNATURE-----  NAME: S.Narendra. GM(O&M) SIGNATURE----- 
<u>7. APPROVED BY</u>	SIGNATURE:-----  for GENERAL MANAGER(BUH)
<u>8.DATE OF ISSUE</u>	
<u>9.INDEX LIST</u>	ENCLOSED.
<u>10.DISTRIBUTION LIST</u>	ENCLOSED.

NTPC

SIMHADRI.

LOCATION MANAGEMENT INSTRUCTION

LMI/OGN/OPS/CIV/005/

ISSUE NO.1; REVISION NO. 3.00; DATE: JULY'2012

INSPECTION OF CONCRETE CHIMNEY

APPROVED FOR

S. Narasimha

Implementation-----

for GM(BUH), SIMHADRI.

Date-----

09/12/2012

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2.0 INTRODUCTION

The RCC chimneys of stage I & II are designed and constructed at Simhadri TPS to release the flue gasses of both stages i.e 2 x 500 MW & 2 x 500 MW through MS stacks to the atmosphere at a height of 275 mt. from ground level. The chimney shell is constructed with RCC and external surface is painted with red and white bands of with as per the standard aviation procedure/rules for aviation obstacles making also aviation obstacles lights are provided at top of the chimney.

3. SCOPE

The scope of the LMI is to detail out the methods of instructions which are to be adopted for RCC chimney.

4. REFERENCE DOCUMENTS

Operation guidance note OGN/OPS/CIV/005 circulated by operation services – NTPC Corporate centre.

5. SUPERCEDED DOCUMENTS

LMI/OGN/OPS/CIV/005 rev.02 dt. July '10

6. SPECIFIC INSTRUCTIONS

The following aspects are to be focused to carry out effective inspection of chimneys by the committee constituted by GM (O&M) which shall include members from Boiler Maint, Elec.Maint, Civil Maint., MTP and FQA

- a) Visual inspection is to be carried out on RCC shell for carbonates, sulphate attacks, chloride attacks and physical and mechanical damages.
- b) Visual inspections for damages of platforms structural staircases, expansion joints
- c) Inspections are to be carried out on electrical installations.
- d) Inspections are to be carried out for safety elevators.
- e) Inspection are to be carried out on aviation obstructing installations
- f) Inspection is to be carried out on steel shell

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LMI

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TITLE: INSPECTION OF CHIMNEY

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7.0 METHODOLOGY OF IMPLEMENTATION & RESPONSIBILITY CENTERS

I) RCC CHIMNEY

A) FOUNDATION

Inspection is to be done for any possible settlement of foundation of chimney with leveling instrument once in a year with reference to the permanent bench mark.

-----By Inspection committee

B) RCC SHELL

1) The outer shell which is constructed with RCC is to be inspected once in a year for the following defects.

a) Cracks: Visual inspection is to be carried out on the outer surface of shell with the help of a binocular to notice any cracks formed on the RCC surface and any exposure of reinforcement, rusting of reinforcement and condition of painting on outer surface.

-----BY Inspection committee

b) Visual inspection is to be carried out for any cracks at the opening of the flue ducts due to variation in temperature.

-----BY Inspection committee

c) Visual inspection is to be carried out for the effective functioning and condition of the rolling shutters once in year.

-----BY Inspection committee

d) Stair case: Visual inspection is to be carried out once in a year for corrosion of steel members, loosening of steel members joints, loosening of fixtures bending or buckling of steel members and condition of paint.

-----BY Inspection committee



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- e) Expansion compensators: Visual inspection is to be carried out once in a year.

-----BY Inspection committee

- f) ELECTRICAL: Inspection is to be carried out to check the continuity of earthing, functioning of earth pits for earth resistivity, electrical continuity connection to all levels and location inspection is to be carried out for corrosion of lighting fixtures/arrestors and conductors. Functional checks are to be carried out on distribution boards and lighting panels for continuity of insulation resistivity, functioning of control board circuits, lightness of connections, cleanliness of Panels/MCC once in a year.

-----BY Inspection committee

- g) Aviation obstructions lighting: Inspection is to be carried out for condition of lighting fixtures, functioning of lights, photos cells and controls units. Inspection is to be carried for the tightness of terminals and JB covers once in a month by EM.

- h) Ventilation and pressurization system: Inspection is to be carried out for the functioning of all system, components/accessories including fans, motors, associated electrical ducting, dampers, air washers units, filters.

-----BY Inspection committee

- i) Communication system: Inspection is to be carried out for the condition of connections, telephone, and condition of handsets at various levels one in six months by C&I.
- j) Elevator: Inspection is to be carried once in a year for functioning of lift, load tests are to be carried on the lift and wire ropes and other moving parts of the elevator including motor and gear box.

-----BY Inspection committee



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8. CHECK LIST

The following are the check list and the reporting format.

S.NO	DESCRIPTION OF ITEM	DATE OF CHECK	TYPE OF CHECK	OBSERVATION
A	FOUNDATION		SETTLEMENT	
B	RCC SHELL		CRACKS ON SHELL SURFACE, SPALLING OF CONCRETE & EXPOSURE OF RE-BARS.	
C	FLUE DUCT		CRACKS ON DUCT OPENING SURFACE	
D	ROLLING SHUTTER		RUSTING, MOVABILITY WORKING CONDITION, HOOD	
E	STAIR CASE		CORRISION, LOOSENING OF JOINTS, BENDING, BUCKLING OF STEEL MEMBERS, CONDITION OF ANCHOR BOLTS.	
F	EXPANSION COMPENSATORS		CONDITION AND DUST ACCUMULATION EARTH.	
G	ELECTRICAL AVIATION		ELECRICAL CONTINUITY	
H	LIGHTING		FUNCTIONING OF LIGHTS	
I	VENTILATION AND PRESSURISATION ON SYSTEM		AT ALL FLOORS	
J	COMMUNICATION SYSTEM		CONDITION OF INSTRUMENT/CONNECTION	
K	ELEVATOR		LOAD TESTS, PHYSICAL OPERATING CONDITIONS W.R.T MECH & ELEC PROTECTION & INTERLOCKING CHECKS FOR SAFETY, VARIFY MAINT LOG BOOK FOR MAJOR REPAIR/REPLACEMENTS.	



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9. EXCEPTION REPORTING

The exceptions encountered during inspection are to be reported as per the following committee.

-----By inspection committee

Description of exception	Stage I / II	Location of exception	Remarks

10. RESPONSIBILITIES

The responsibility of carrying out different tasks in relation to inspection of concrete chimney as detailed in the LMI are with the inspection committee.

11. RECORD KEEPING:

The records to be maintained under this LMI include

- Compliance record of the activities detailed under this LMI.
- Reports of inspection carried out.
- Exception reporting.

The above records are to be maintained by O&M-Civil, EMD, EM for the relevant responsibilities.

12. REVIEW:

The document shall be reviewed by HOP on two yearly basis.

